

High-Frequency Vibrational Power Flows in Randomly Heterogeneous Structures

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ABSTRACT: We derive the evolution properties of the high-frequency vibrational energy density in slender visco-elastic structures, taking the example of beams and thick shells. For that purpose we apply some recent mathematical results which show that this quantity satisfies a transport equation, or a so-called radiative transfer equation in the presence of random heterogeneities. The latter can be approached by a diffusion equation after long propagation times; we also derive the diffusive characteristics of beams and shells. Some numerical simulations are presented, using either the discontinuous finite element method or the direct Monte-Carlo method to solve the radiative transfer equations. Comparisons are made with the analytical solution of a diffusion equation and we discuss the relevance of the vibrational conductivity analogy of high-frequency structural-acoustics invoked by some researchers.

1 INTRODUCTION

Two approaches are currently used by engineers to predict the high-frequency vibrational behavior of elastic structures. Statistical energy analysis, known as SEA (Lyon and DeJong 1995), may be called a global approach in the sense that it gives estimates of the mean vibrational energy within substructures of a complex mechanical system. A major difficulty of the method, which is still very heuristic, is the determination of the physical parameters which contribute to the formulation: coupling loss factors, modal densities, input powers. The vibrational conductivity analogy of structural-acoustics, or power flow analysis (Nefske and Sung 1989; Bouthier and Bernhard 1992; Langley 1995; Lase et al. 1996), may be called a local model because it gives estimates of the vibrational energy and power flow densities. However its developments to date are restricted to simple homogeneous structures (beams or more rarely plates) because it is based on some very restrictive hypotheses which are difficult to fulfil – or are simply wrong – for more complex structures.

The modern mathematical theory of microlocal analysis (Lions and Paul 1993; Gérard et al. 1997; Papanicolaou and Ryzhik 1999) shows that in fact the energy density associated to high-frequency waves satisfy Liouville-type transport equations, or radiative transfer equations for randomly heterogeneous media. Only after long propagation times or for short scat-

tering mean free paths can the latter be approached by diffusion equations. The analysis is based on the use of a Wigner transform and an explicit scaling of wave propagation patterns. The high-frequency non-negative limit of the Wigner transform is the angularly resolved energy density in time and space. This quantity also characterizes the power flows within the medium, thus allowing us to track the energy paths. The purpose of this communication is to focus on such a transport theory for isotropic visco-elastic slender structures such as beams or shells (Savin 2004).

First, we derive the transport/radiative transfer equations adapted to these systems, extending the existing mathematical results to visco-elastic materials with memory effects. In order to obtain a radiative transfer regime, materials constituting the structures are assumed to be randomly heterogeneous at scales comparable to the wavelengths of the excitation. This is the *a priori* worst configuration since we can expect strong interactions between waves and heterogeneities of similar sizes. The density, shear modulus, and Young's modulus are modeled by statistically homogeneous random processes. The sole quantities necessary to solve the radiative transfer equations are their auto- and cross-power spectral density functions. Then these equations are solved numerically either by a discontinuous Galerkin finite element method, as done for the example of a curved Timoshenko beam, or by a direct Monte-Carlo method, as done for the

example of a thick cylindrical shell. We finally obtain the diffusion properties of these structures, namely the scattering and transport mean free paths as well as the diffusion constants. The final section of the paper is dedicated to a short discussion on the results obtained and the choice of the numerical scheme.

2 WIGNER MEASURE AS ENERGY DENSITY ESTIMATE

We summarize in this section the basic developments of (Gérard et al. 1997; Papanicolaou and Ryzhik 1999; Akian 2003). Consider a first-order hyperbolic system of the form

$$\mathbf{A}(\mathbf{x}, 0) \partial_t \mathbf{v} + \sum_{j=1}^d \mathbf{D}_j \partial_{x_j} (\mathbf{v} + \mathbf{B}(\mathbf{x}, \cdot) *_t \mathbf{v}) + \mathbf{C} \mathbf{v} = \mathbf{0} \quad (1)$$

where $\mathbf{v}$ is the vector of state variables in $\mathbb{C}^n$, and $\mathbf{x} \in \mathbb{R}^d$ runs through the physical space; $\mathbf{A}(\mathbf{x}, t)$ is a symmetric, positive definite $n \times n$ real matrix; $\mathbf{D}_j$ are symmetric $n \times n$ real matrices independent of position $\mathbf{x}$ and time t ; $\mathbf{B}(\mathbf{x}, t)$ is a symmetric $n \times n$ real matrix; $\mathbf{C}(\mathbf{x}, t)$ is a skew-symmetric $n \times n$ real matrix; and $*_t$ is the time convolution product $f *_t g(t) = \int_0^t f(\tau) g(t - \tau) d\tau$, $t \geq 0$. We denote by $\mathbf{v}^\epsilon$ the solution of Eq. (1) for initial conditions oscillating rapidly (in the sense precised below) at an arbitrarily small scale ϵ , considering slowly varying mechanical coefficients such that neither $\mathbf{A}$ nor $\mathbf{B}$ depend on ϵ . Note that the setting of Eq. (1) is very general for wave equations arising from physical problems: it embeds Maxwell's equations for electromagnetic waves, the wave equation for acoustics or the Navier equation for three-dimensional elastodynamics (Papanicolaou and Ryzhik 1999).

2.1 General Considerations

We inject high-frequency energy into the system described by Eq. (1) through, say, initial conditions of the form

$$\mathbf{v}^\epsilon(\mathbf{x}, 0) = \mathbf{v}_0^\epsilon(\mathbf{x}), \quad (2)$$

where the data $\mathbf{v}_0^\epsilon$ are oscillating at most at the frequency $\frac{1}{\epsilon}$. The mechanical energy density associated to the solution $\mathbf{v}^\epsilon$ of eqs. (1)–(2) is defined by

$$\mathcal{E}^\epsilon(\mathbf{x}, t) = \frac{1}{2} (\mathbf{v}^\epsilon, \mathbf{v}^\epsilon)_{\mathbf{A}(\mathbf{x}, 0)} \quad (3)$$

while the densities of power flow along each direction $j \in \{1, 2, \dots, d\}$ are

$$\Pi_j^\epsilon(\mathbf{x}, t) = \frac{1}{2} (\mathbf{v}^\epsilon, \mathbf{v}^\epsilon)_{\mathbf{D}_j} \quad (4)$$

with the notation $(\mathbf{u}, \mathbf{v})_{\mathbf{M}} = (\mathbf{M}\mathbf{u}, \mathbf{v})$. A potential good candidate to study the high-frequency energetic features of the system (1)–(2) is the Wigner transform,

because it has a direct link with the above quantities in the limit $\epsilon \rightarrow 0$, as well as some other nice evolution properties given below. The $n \times n$ matrix-valued Wigner transform $\mathbf{W}^\epsilon[\mathbf{v}]$ of a function $\mathbf{v}$ with finite energy is defined by:

$$\mathbf{W}^\epsilon[\mathbf{v}](\mathbf{x}, \mathbf{k}, t) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{y}} \mathbf{v} \left(\mathbf{x} - \frac{1}{2} \epsilon \mathbf{y}, t \right) \mathbf{v}^* \left(\mathbf{x} + \frac{1}{2} \epsilon \mathbf{y}, t \right) d\mathbf{y} \quad (5)$$

where $\mathbf{v}^*$ stands for the conjugate transpose. Then $\mathbf{W}^\epsilon[\mathbf{v}^\epsilon]$ for $\mathbf{v}^\epsilon$ of (1)–(2) given above has an hermitian weak limit (in the sense of tempered distributions) $\mathbf{W}^\epsilon[\mathbf{v}^\epsilon] \rightharpoonup \mathbf{W}$ as $\epsilon \rightarrow 0$, the so-called Wigner measure. We assume that $\mathbf{v}^\epsilon$ has the same oscillatory properties as the initial conditions, then it can be shown that the energy $\mathcal{E}(\mathbf{x}, t)$ and power flow $\Pi_j(\mathbf{x}, t)$ densities in the limit $\epsilon \rightarrow 0$ are estimated from its Wigner measure by:

$$\mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \int_{\mathbb{R}^d} \mathbf{A}(\mathbf{x}, 0) : \mathbf{W}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} \quad (6)$$

and for $j \in \{1, 2, \dots, d\}$:

$$\Pi_j(\mathbf{x}, t) = \frac{1}{2} \int_{\mathbb{R}^d} \mathbf{D}_j : \mathbf{W}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad (7)$$

respectively, where $\mathbf{M} : \mathbf{N} = \text{Tr}(\mathbf{M}\mathbf{N}^T)$ stands for the usual tensor scalar product and Tr is the trace of a matrix.

We now turn to the evolution properties of the Wigner measure. Let:

$$\mathbf{L}(\mathbf{x}, \mathbf{k}) = \mathbf{A}^{-1}(\mathbf{x}, 0) \sum_{j=1}^d k_j \mathbf{D}_j. \quad (8)$$

This is the dispersion matrix of the above system, which is self-adjoint with respect to the scalar product $(\mathbf{u}, \mathbf{v})_{\mathbf{A}(\mathbf{x}, 0)}$. Thus its n right eigenvectors form an orthonormal family with respect to this scalar product. The latter and the associated real eigenvalues are denoted by $\mathbf{b}_{\alpha r}(\mathbf{x}, \mathbf{k})$ and $\lambda_{\alpha r}(\mathbf{x}, \mathbf{k})$, respectively, for $1 \leq \alpha \leq M \leq n$ and $1 \leq r \leq R_\alpha$, where R_α is the order of multiplicity of eigenvalue λ_α for a mode α among the M existing ones, such that $\sum_{\alpha=1}^M R_\alpha = n$. We assume that R_α does not depend on $(\mathbf{x}, \mathbf{k})$. Then it can be shown that the Wigner measure writes:

$$\mathbf{W}(\mathbf{x}, \mathbf{k}, t) = \sum_{\alpha=1}^M \sum_{r,s=1}^{R_\alpha} w_{\alpha}^{rs}(\mathbf{x}, \mathbf{k}, t) \mathbf{b}_{\alpha r}(\mathbf{x}, \mathbf{k}) \mathbf{b}_{\alpha s}^*(\mathbf{x}, \mathbf{k})$$

and if we define coherence matrices $\mathbf{W}_\alpha$, $1 \leq \alpha \leq M$, by:

$$W_{\alpha,rs}(\mathbf{x}, \mathbf{k}, t) = w_{\alpha}^{rs}(\mathbf{x}, \mathbf{k}, t), \quad 1 \leq r, s \leq R_\alpha,$$

then they satisfy the following transport equations:

$$\partial_t \mathbf{W}_\alpha + \{\lambda_\alpha, \mathbf{W}_\alpha\} = \mathbf{0}. \quad (9)$$

$\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ stands for the usual Poisson's bracket. It is important to note at this stage that matrix $\mathbf{B}$ (accounting for the material viscous memory effects) has no influence on the transport properties of the energy, as proved rigorously in (Akian 2003). Thus these effects are ignored in the remaining of the paper setting $\mathbf{B} \equiv \mathbf{0}$.

2.2 Transport of Energy in a Curved Timoshenko Beam

The axial motion of the beam is denoted by u , its bending deflection by v , its bending angle by θ , and its curvature by R . The mass per unit length is ρ , E is the Young's modulus, κG is the reduced shear modulus where κ is the usual shear reduction factor depending on the cross-section geometry and Poisson's coefficient ν , and $r = \sqrt{I/S}$ is the radius of gyration where I and S are respectively the cross-section inertia and area. The shear force T , axial force N and bending moment M are given by:

$$\begin{cases} T = \kappa G S (\partial_x v - \theta), \\ N = E S (\partial_x u + \frac{v}{R}), \\ M = E I \partial_x \theta \end{cases}$$

where x is the coordinate along the beam. The dynamic equilibrium equations are:

$$\begin{cases} \rho \partial_t^2 u = \partial_x N + \frac{M}{R}, \\ \rho \partial_t^2 v = \partial_x T - \frac{N}{R}, \\ \rho r^2 \partial_t^2 \theta = T + \partial_x M. \end{cases}$$

Let $\mathbf{u} = (\partial_t u, \partial_t v, \partial_t \theta, N, T, M)$, then these equations take the form of Eq. (1) with $d = 1$, $n = 6$, $\mathbf{A}(x) = \text{diag}\{\rho, \rho, \rho r^2, 1/ES, 1/\kappa GS, 1/EI\}$, and:

$$\mathbf{C} = \begin{bmatrix} 0 & \mathbf{C}_3 \\ -\mathbf{C}_3^T & 0 \end{bmatrix}, \quad \mathbf{C}_3 = \begin{bmatrix} 0 & 0 & -\frac{1}{R} \\ \frac{1}{R} & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}.$$

The expression of matrix $\mathbf{D}$ is not reproduced here as it may be easily deduced from the previous set of equations. The dispersion matrix $\mathbf{L}$ is then:

$$-\mathbf{L}(x, k) = \begin{bmatrix} 0 & 0 & 0 & \frac{k}{\rho} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{k}{\rho} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{k}{\rho r^2} \\ kES & 0 & 0 & 0 & 0 & 0 \\ 0 & k\kappa GS & 0 & 0 & 0 & 0 \\ 0 & 0 & kEI & 0 & 0 & 0 \end{bmatrix}.$$

Its eigenvalues on the set $\mathbb{R}^2 \setminus \{(x, k); k = 0\}$ are given by:

$$\begin{aligned} \lambda_{\mp}^{\pm} &= \pm c_T |k| \quad \text{each with multiplicity 1,} \\ \lambda_{\mp}^{\pm} &= \pm c_P |k| \quad \text{each with multiplicity 2,} \end{aligned} \quad (10)$$

where $c_T^2 = \kappa GS/\rho$ and $c_P^2 = ES/\rho$, and the associated eigenvectors are:

$$\begin{aligned} \mathbf{b}_{\mp}^{\pm} &= \left(0, \frac{\text{sgn}(k)}{\sqrt{2\rho}}, 0, 0, \mp \sqrt{\frac{\kappa Gh}{2}}, 0 \right), \\ \mathbf{b}_{\text{Pb}}^{\pm} &= \left(0, 0, \frac{\text{sgn}(k)}{\sqrt{2\rho r^2}}, 0, 0, \mp \sqrt{\frac{EI}{2}} \right), \\ \mathbf{b}_{\text{Pn}}^{\pm} &= \left(\frac{\text{sgn}(k)}{\sqrt{2\rho}}, 0, 0, \mp \sqrt{\frac{ES}{2}}, 0, 0 \right), \end{aligned} \quad (11)$$

with sgn denoting the sign function. The scalar transport equation for shear energy is then:

$$\partial_t w_{\mp}^{\pm} \pm \text{sgn}(k) c_T \partial_x w_{\mp}^{\pm} \mp |k| \partial_x c_T \cdot \partial_k w_{\mp}^{\pm} = 0, \quad (12)$$

and the matrix-valued transport equation for the axial and bending energy modes (having the same phase speed) is:

$$\partial_t \mathbf{W}_{\text{P}}^{\pm} \pm \text{sgn}(k) c_P \partial_x \mathbf{W}_{\text{P}}^{\pm} \mp |k| \partial_x c_P \cdot \partial_k \mathbf{W}_{\text{P}}^{\pm} = \mathbf{0}. \quad (13)$$

Thus the forward ('+') and backward ('-') energy waves satisfy $w_{\mp}(x, k, t) = w_{\mp}^{\pm}(x, -k, t)$ and $\mathbf{W}_{\text{P}}^-(x, k, t) = \mathbf{W}_{\text{P}}^+(x, -k, t)$. The total energy density is given by:

$$\mathcal{E}(x, t) = \int_{\mathbb{R}} w_{\mp}^+(x, k, t) dk + \int_{\mathbb{R}} \text{Tr} \mathbf{W}_{\text{P}}^+(x, k, t) dk \quad (14)$$

and the power flow density (or structural intensity) is given by:

$$\begin{aligned} \Pi(x, t) &= c_T(x) \int_{\mathbb{R}} \text{sgn}(k) w_{\mp}^+(x, k, t) dk \\ &+ c_P(x) \int_{\mathbb{R}} \text{sgn}(k) \text{Tr} \mathbf{W}_{\text{P}}^+(x, k, t) dk. \end{aligned} \quad (15)$$

2.3 Transport of Energy in a Cylindrical Shell

Transport properties of the high-frequency energy in a thick cylindrical shell have been derived elsewhere (Savin 2004) and will not be reproduced here. We just remind that for a shell of thickness h , Young's modulus E , reduced shear modulus κG , Poisson's coefficient ν and mass per unit surface ρ , three modes of energy transport can be found: a pure shear mode of multiplicity one associated to the phase speed $c_T^2 = \kappa Gh/\rho$, and two coupled in-plane/bending modes each of multiplicity two associated to the phase speeds $c_S^2 = Gh/\rho$ and $c_P^2 = Eh/(1 - \nu^2)\rho$, respectively.

3 INFLUENCE OF RANDOM HETEROGENEITIES: RADIATIVE TRANSFER

3.1 General Considerations

In this section we consider some randomness of the mechanical properties of the medium and follow again the developments of (Papanicolaou and Ryzhik 1999). Let us introduce a random perturbation $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ of matrix $\mathbf{A}(\mathbf{x}, 0)$ of the form:

$$\mathbf{A}(\mathbf{x}, 0) = \underline{\mathbf{A}}(\mathbf{x}, 0) \left[\mathbf{I}_n + \sqrt{\epsilon} \mathbf{V} \left(\frac{\mathbf{x}}{\epsilon} \right) \right], \quad (16)$$

where $\mathbf{I}_n$ stands for the $n \times n$ identity matrix. The magnitude $\sqrt{\epsilon}$ of perturbations is chosen such that they have comparable effects to the unperturbed background medium characterized by the mean matrix of parameters $\underline{\mathbf{A}}(\mathbf{x}, 0)$. However they need not be too strong, for in this case localization rather than transport may occur. Their correlation lengths are chosen to be comparable to wavelengths ϵ to allow strong interactions between the waves and the heterogeneities. In order that the system (1) remains symmetric hyperbolic, the perturbation matrix has to satisfy the condition $\underline{\mathbf{A}}(\mathbf{x}, 0) \mathbf{V}(\mathbf{y}) = \mathbf{V}^*(\mathbf{y}) \underline{\mathbf{A}}(\mathbf{x}, 0)$ for all $\mathbf{x}$ and $\mathbf{y}$ of $\mathbb{R}^d$. Parameter fluctuations are modeled by a statistically homogeneous, second order matrix-valued random process with mean zero whose covariance functions are denoted by $R_{jklm}(\mathbf{x})$ and such that:

$$\begin{aligned} R_{jklm}(\mathbf{x} - \mathbf{y}) &= \mathbb{E} \{ V_{jk}(\mathbf{x}) V_{lm}(\mathbf{y}) \} \\ &= \int_{\mathbb{R}^d} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} S_{jklm}(\mathbf{k}) d\mathbf{k} \end{aligned}$$

where $\mathbb{E}\{\cdot\}$ stands for mathematical expectation. $\mathbf{k} \mapsto S_{jklm}(\mathbf{k})$ are the spectral density functions of random matrix $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ for $j, k, l, m \in \{1, 2, \dots, n\}$ and let $\mathbf{S}$ be the fourth-order spectral density tensor corresponding to these functions. Then it can be shown that the coherence matrices $\mathbf{W}_\alpha, 1 \leq \alpha \leq M$, satisfy coupled radiative transfer equations given by:

$$\begin{aligned} \partial_t \mathbf{W}_\alpha + \{\lambda_\alpha, \mathbf{W}_\alpha\} &= \\ \sum_{\beta=1}^M \int_{\mathbb{R}^d} \sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') (\mathbf{W}_\beta(\mathbf{k}') - \mathbf{W}_\alpha(\mathbf{k})) d\mathbf{k}'. \end{aligned} \quad (17)$$

The differential scattering operators $\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}')$ are defined by:

$$\begin{aligned} \sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_\beta(\mathbf{k}') &= 2\pi \lambda_\alpha^2(\mathbf{k}) \delta(\lambda_\alpha(\mathbf{k}) - \lambda_\beta(\mathbf{k}')) \\ &\quad \times \mathcal{S}_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_\beta(\mathbf{k}') \end{aligned} \quad (18)$$

where $\mathcal{S}_{\alpha\beta}(\mathbf{k}, \mathbf{k}')$ is a fourth-order tensor with elements

$$\mathcal{S}_{rstu}^{\alpha\beta}(\mathbf{k}, \mathbf{k}') = \mathbf{c}_{\alpha r}(\mathbf{k}) \mathbf{b}_{\beta t}^*(\mathbf{k}') : \mathbf{S}(\mathbf{k} - \mathbf{k}') : \mathbf{c}_{\alpha s}(\mathbf{k}) \mathbf{b}_{\beta u}^*(\mathbf{k}').$$

$\mathbf{b}_{\alpha r}$ are the right eigenvectors of the mean dispersion matrix and $\mathbf{c}_{\alpha r}$ are its left eigenvectors, given by $\mathbf{c}_{\alpha r} = \underline{\mathbf{A}} \mathbf{b}_{\alpha r}$, for $1 \leq \alpha \leq M$ and $1 \leq r \leq R_\alpha$. The scattering cross-section $\sigma_{\alpha\beta}$ gives the rate at which the energy mode β with wave vector $\mathbf{k}'$ is converted to the energy mode α with wave vector $\mathbf{k}$, at position $\mathbf{x}$. The total scattering cross-section Σ_α such that:

$$\Sigma_\alpha(\mathbf{k}) = \sum_{\beta=1}^M \int_{\mathbb{R}^d} \sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{I}_{R_\beta} d\mathbf{k}'$$

gives the rate at which the energy mode α with wave vector $\mathbf{k}$ at position $\mathbf{x}$ is converted to all other modes or wave numbers.

3.2 Radiative Transfer in a Timoshenko Beam

We consider for a Timoshenko beam in bending a random matrix $(\mathbf{V}(x), x \in \mathbb{R})$ which takes the form:

$$\mathbf{V}(x) = \text{diag} \left\{ \tilde{\rho}(x) \mathbf{I}_3, \tilde{E}(x), \tilde{G}(x), \tilde{E}(x) \right\} \quad (19)$$

where $(\tilde{\rho}(x), x \in \mathbb{R})$ (for density fluctuations), $(\tilde{G}(x), x \in \mathbb{R})$ (inverse reduced shear modulus fluctuations) and $(\tilde{E}(x), x \in \mathbb{R})$ (inverse Young's modulus fluctuations) are three real-valued, mean-zero, statistically homogeneous stochastic processes of which spectral measures are characterized by their power spectral density functions $k \mapsto S_{\rho\rho}(k)$, $k \mapsto S_{GG}(k)$ and $k \mapsto S_{EE}(k)$, respectively, and cross-spectral density functions $k \mapsto S_{\rho G}(k)$ and $k \mapsto S_{\rho E}(k)$. The mean matrix of parameters is $\underline{\mathbf{A}}(x) = \text{diag} \{ \varrho, \varrho, \varrho^2, 1/ES, 1/\kappa GS, 1/EI \}$; matrix $\mathbf{V}$ obviously satisfies the commutative condition $\underline{\mathbf{A}}(x) \mathbf{V}(y) = \mathbf{V}^*(y) \underline{\mathbf{A}}(x)$ and matrix $\mathbf{A}(x) = \underline{\mathbf{A}}(x) [\mathbf{I}_6 + \sqrt{\epsilon} \mathbf{V}(\frac{x}{\epsilon})]$ remains symmetric, positive definite. The differential scattering cross-sections are then:

$$\begin{aligned} \sigma_{\text{PP}}^+(k, k') &= \sigma_{\text{PP}}^-(k, k') := \sigma_{\text{PP}}(k, k') \\ \sigma_{\text{TT}}^+(k, k') &= \sigma_{\text{TT}}^-(k, k') := \sigma_{\text{TT}}(k, k') \end{aligned}$$

with

$$\sigma_{\text{PP}}(k, k') = \frac{\pi}{2} c_p^2 k^2 [S_{\rho\rho}(k - k') + S_{EE}(k - k')$$

$$+ 2\text{sgn}(k) \text{sgn}(k') \Re S_{\rho E}(k - k')] \delta(c_p |k| - c_p |k'|)$$

and

$$\sigma_{\text{TT}}(k, k') = \frac{\pi}{2} c_T^2 k^2 [S_{\rho\rho}(k - k') + S_{GG}(k - k')$$

$$+ 2\text{sgn}(k) \text{sgn}(k') \Re S_{\rho G}(k - k')] \delta(c_T |k| - c_T |k'|),$$

for $\Re(z)$ being the real part of a complex z . All other cross-sections are identically null. The radiative transfer equations are:

$$\partial_t \mathbf{W}_p^+ + \{\lambda_p^+, \mathbf{W}_p^+\} = \frac{\pi}{2} c_p k^2 [S_{\varrho\varrho}(2k) + S_{EE}(2k) - 2\Re S_{\varrho E}(2k)] (\mathbf{W}_p^+(-k) - \mathbf{W}_p^+(k)) \quad (20)$$

for the axial and bending motions,

$$\partial_t w_T^+ + \{\lambda_T^+, w_T^+\} = \frac{\pi}{2} c_T k^2 [S_{\varrho\varrho}(2k) + S_{GG}(2k) - 2\Re S_{\varrho G}(2k)] (w_T^+(-k) - w_T^+(k)) \quad (21)$$

for pure shear, and the symmetric for $\mathbf{W}_p^-$ and w_T^- .

3.3 Radiative Transfer in a Cylindrical Shell

The radiative transfer regime in a thick cylindrical shell has been derived elsewhere (Savin 2004) and, again, will not be described here. The scattering cross-sections are similar to the ones obtained previously for a Mindlin plate (Savin 2002).

4 NUMERICAL SOLUTION OF RADIATIVE TRANSFER EQUATIONS

Radiative transfer equations can be solved numerically by various methods, see for example (Duderstadt and Martin 1979; Dautray and Lions 1985). For the case of a Timoshenko beam, we use the discontinuous Galerkin finite element method originally developed for neutronic transport (Reed and Hill 1973; Lesaint and Raviart 1974). For cylindrical shells, we have used a direct Monte-Carlo method (Lapeyre et al. 1998).

4.1 Discontinuous Galerkin Finite Element Method for Beams

We consider an homogeneous background medium and a scalar radiative transfer equation such that the latter writes (see Eq. (12) and Eq. (21)):

$$\frac{1}{c_\alpha} \partial_t w_\alpha + \text{sgn}(k) \partial_x w_\alpha = \tilde{\sigma}_\alpha(k) (w_\alpha(-k) - w_\alpha(k)) \quad (22)$$

for either $\alpha = T$ or $\alpha = P$ for which we define $w_p = \text{Tr} \mathbf{W}_p^+$. We solve it numerically in the phase space $\Omega = [a, b] \times \mathbb{R}$ with, say, specular reflection conditions and no incoming flux on the boundaries $x = a$ and $x = b$:

$$w_\alpha(a, k, t) = w_\alpha(a, -k, t), \quad w_\alpha(b, -k, t) = w_\alpha(b, k, t),$$

and a known initial condition $w_\alpha(x, k, 0) = w_0(x, k)$. Let us introduce the space of allowable solutions $V =$

$H^1(\Omega)$, then the variational formulation of this problem writes: Find $w_\alpha \in V$ with values in $\mathbb{R}_+$ such that:

$$\begin{aligned} & \frac{1}{c_\alpha} \int_\Omega \partial_t w_\alpha(x, k, t) v(x, k) \, dx dk \\ & - \int_\Omega \text{sgn}(k) w_\alpha(x, k, t) \partial_x v(x, k) \, dx dk \\ & + \int_\Omega \tilde{\sigma}_\alpha(k) [w_\alpha(x, k, t) - w_\alpha(x, -k, t)] v(x, k) \, dx dk \\ & + \int_{-\infty}^0 w_\alpha(a, k, t) [v(a, k) - v(a, -k)] \, dk \\ & + \int_0^{+\infty} w_\alpha(b, k, t) [v(b, k) - v(b, -k)] \, dk = 0 \end{aligned}$$

for all $v \in V$. The solution is continuous with respect to the space variable x although it may present strong gradients at some points. It is however discontinuous with respect to the phase variable k , especially at $k = 0$. That is the reason why discontinuous finite element methods have been developed in order to properly take into account such discontinuities or quasi-discontinuities. If we denote by x_D a point of spatial quasi-discontinuity, then the above weak variational formulation is modified to:

$$\begin{aligned} & \frac{1}{c_\alpha} \int_\Omega \partial_t w_\alpha(x, k, t) v(x, k) \, dx dk \\ & - \int_\Omega \text{sgn}(k) w_\alpha(x, k, t) \partial_x v(x, k) \, dx dk \\ & + \int_\Omega \tilde{\sigma}_\alpha(k) [w_\alpha(x, k, t) - w_\alpha(x, -k, t)] v(x, k) \, dx dk \\ & + \int_{-\infty}^0 w_\alpha(a, k, t) [v(a, k) - v(a, -k)] \, dk \\ & + \int_0^{+\infty} w_\alpha(b, k, t) [v(b, k) - v(b, -k)] \, dk \\ & + \int_{-\infty}^0 w_\alpha(x_D^+, k, t) [v(x_D^+, k) - v(x_D^-, k)] \, dk \\ & + \int_0^{+\infty} w_\alpha(x_D^-, k, t) [v(x_D^-, k) - v(x_D^+, k)] \, dk = 0 \end{aligned} \quad (23)$$

for all $v \in V$. The continuity of the solution at $x = x_D$ is ensured by the conditions

$w_\alpha(x_D^+, k, t) = w_\alpha(x_D^-, k, t)$ if $k \geq 0$ (forward flux) and $w_\alpha(x_D^-, k, t) = w_\alpha(x_D^+, k, t)$ if $k \leq 0$ (backward flux).

The phase space is partitioned into non-overlapping sub-domains, or elements, $\mathcal{D}_i$ and an approximate solution w_α^h to this problem is sought in a finite element subspace $V_h = \{v \in L^1(\Omega); v|_{\mathcal{D}_i} \in \mathcal{P}(\mathcal{D}_i)\}$, where $\mathcal{P}(\mathcal{D}_i)$ is a polynomial space defined on $\mathcal{D}_i$. Local basis sets $\{\phi_l^i\}_{l=0,1,\dots,N_i-1}$ such that $\mathcal{P}(\mathcal{D}_i) = \text{span}(\{\phi_l^i\}_{l=0,1,\dots,N_i-1})$ and $\text{supp}(\phi_l^i(x, k)) = \mathcal{D}_i$ are introduced on each element, and the approximate solution is written:

$$w_\alpha^h|_{\mathcal{D}_i}(x, k, t) = \sum_{l=0}^{N_i-1} \theta_l^i(t) \phi_l^i(x, k).$$

The above Eq. (23) then leads to a system of the form (Duderstadt and Martin 1979):

$$\frac{1}{c_\alpha} \mathbf{M} \dot{\boldsymbol{\theta}} + \mathbf{K} \boldsymbol{\theta} = \mathbf{0} \quad (24)$$

where $\boldsymbol{\theta}$ is the vector of the degrees of freedom θ_l^i .

This procedure has been applied to a beam $b = -a = 10$ partitioned into 400 space elements and 2 phase elements. The initial condition is:

$$w_0(x, k) = e^{-\ln 2(x/r_0)^2} \otimes \delta(k-1)$$

and the covariance functions of parameters fluctuations are chosen as (Gaussian model):

$$R_{pq}(x-y) = r_{pq} e^{-\gamma^2|x-y|^2}, S_{pq}(k) = \frac{r_{pq}}{2\sqrt{\pi}\gamma} e^{-\frac{k^2}{4\gamma^2}}$$

for $p, q \in \{\varrho, E, G\}$. γ^{-1} is their correlation length and the r_{pq} 's are scaling parameters such that $|r_{pq}| \leq 1$. Legendre polynomials up to the second order are used as local basis functions on uniform elements for the space dimension, and Laguerre polynomials up to the sixth order are used for the wave number dimension. Then Eq. (24) is solved in time by a fourth-order Runge-Kutta scheme. Fig. (1) displays our results for the data $r_0 = 0.5$, $\gamma^{-1} = 2$, $r_{\varrho\varrho} = r_{EE} = r_{GG} = 1$, and $r_{\varrho E} = r_{\varrho G} = 0$. The time scale is $T = b/c_\alpha$, the time needed by the wave front to reach the extremity of the beam. A small amount of damping $\eta_\alpha = 0.01$ has been added such that the displayed solution is the approximate energy density

$$\mathcal{E}_\alpha^h(x, t) = e^{-\eta_\alpha t} \int_{\mathbb{R}} w_\alpha^h(x, k, t) dk.$$

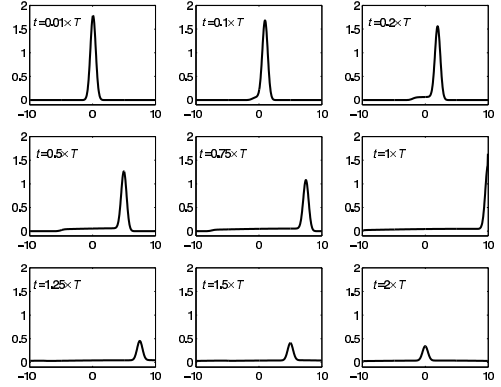


Figure 1: High-frequency energy propagation in a Timoshenko beam with Gaussian correlations of parameters.

4.2 Direct Monte-Carlo Method for Cylindrical Shells

The direct Monte-Carlo method consists in simulating particle trajectories of a particular class of transport processes and then averaging over these paths. Again, we consider the case of an homogeneous constant-velocity background medium with statistically isotropic random heterogeneities. The system of radiative transfer equations for in-plane/bending and pure shear energy propagation in a thick shell ($d = 2$ with the notations of section 2) is:

$$\partial_t \mathbf{w} + \mathbf{c} \hat{\mathbf{k}} \cdot \nabla \mathbf{w} + \Sigma(\mathbf{k}) \mathbf{w} =$$

$$\int_{S^1} \tilde{\sigma}(|\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \mathbf{w}(\mathbf{x}, \mathbf{k}', t) d\hat{\mathbf{k}}' \quad (25)$$

where $\mathbf{c} = \text{diag}(c_T, c_p, c_s)$ is the velocity matrix, $\Sigma = \text{diag}(\Sigma_T, \Sigma_p, \Sigma_s)$ is the total scattering matrix and $\tilde{\sigma}$ is the kernel of the collision operator:

$$\tilde{\sigma} = \begin{bmatrix} \tilde{\sigma}_{TT} & 0 & 0 \\ 0 & \tilde{\sigma}_{pp} & \tilde{\sigma}_{ps} \\ 0 & \tilde{\sigma}_{sp} & \tilde{\sigma}_{ss} \end{bmatrix}. \quad (26)$$

We have defined $\mathbf{w} = (w_T, w_p, w_s)$ with $w_p = \text{Tr} \mathbf{W}_p^+$ and $w_s = \text{Tr} \mathbf{W}_s^+$. At last S^1 stands for the unit circle of $\mathbb{R}^2$ and $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$. The circular cylindrical shell has a length L with, as for the beam, specular reflection conditions at the boundaries $x_1 = \pm L/2$ where we also neglect the possible mode conversions. Fig. (2) displays our results for bending energetic waves with Gaussian correlation of the perturbations:

$$R_{pq}(\mathbf{x} - \mathbf{y}) = r_{pq} e^{-\gamma^2|\mathbf{x} - \mathbf{y}|^2}, S_{pq}(\mathbf{k}) = \frac{r_{pq}}{4\pi\gamma^2} e^{-\frac{|\mathbf{k}|^2}{4\gamma^2}},$$

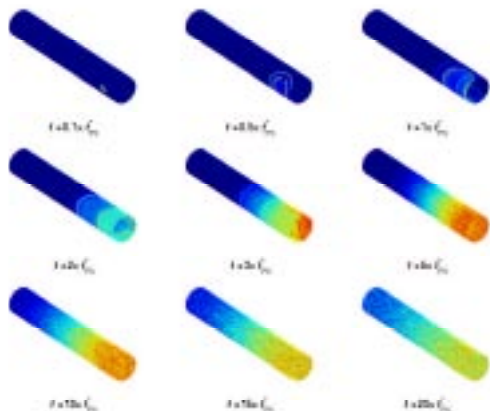


Figure 2: High-frequency energy propagation in a circular cylindrical shell with Gaussian correlations of parameters and $|\mathbf{k}|/\gamma = 1$. After (Savin 2004).

$r_{\varrho\varrho} = 10^{-2}$, $r_{EE} = r_{GG} = 1$, and $r_{\varrho E} = r_{\varrho G} = 0$, $\nu = 0.2$ for Poisson's coefficient, $c_S = 1$ m/s and $|\mathbf{k}|/\gamma = 1$. A unit point source is applied at time $t = 0$ near one extremity of the shaft and it equally loads both bending modes P and S. Time steps are multiples of the diffusion time $t_{PS}^* = \max(\ell_P^*/c_P, \ell_S^*/c_S)$ where the transport mean free paths ℓ_α^* are defined below.

5 DIFFUSIVE REGIME

After long propagation times or for short scattering mean free paths with respect to the structure dimensions, radiative transfer equations can be approximated by diffusion equations. For unbounded, homogeneous media, the scattering mean free paths are infinite and thus no diffusion regime can be reached. This is in contradiction with the usual assumptions of the so-called power flow analysis, or vibrational conductivity analogy, of high-frequency structural-acoustics (Nefske and Sung 1989; Bouthier and Bernhard 1992; Langley 1995; Lase et al. 1996). The above results show that the latter approach is in fact essentially incorrect.

The scattering mean free path characterizes the mean distance a wave must travel in a scattering medium before its direction is significantly altered. As such, it can be interpreted as the mean distance between two consecutive scattering events. It is always lower than or equal to the transport mean path, which is the distance a wave must travel before its direction is completely randomized. For a Timoshenko beam with statistically isotropic (independent of the direction) random heterogeneities, it is given by:

$$\ell_T^*(k) = \frac{1}{\pi k^2 (S_{\varrho\varrho}(2|k|) + S_{GG}(2|k|) - 2\Re S_{\varrho G}(2|k|))}$$

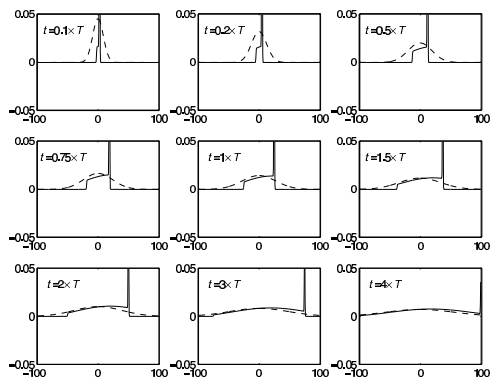


Figure 3: Comparison of radiative transfer (full line) and diffusion (dashed line) solutions for the high-frequency energy density evolution in a Timoshenko beam with Gaussian correlations of parameters and $k_0 = 2$.

for the shear energy density, and by:

$$\ell_P^*(k) = \frac{1}{\pi k^2 (S_{\varrho\varrho}(2|k|) + S_{EE}(2|k|) - 2\Re S_{\varrho E}(2|k|))}$$

for the bending/axial energy density. The diffusion equations are:

$$\partial_t w_\alpha(x, |k|, t) = \partial_x (D_\alpha(|k|) \partial_x w_\alpha(x, |k|, t)) , \quad (27)$$

where the D_α 's are the diffusion constants defined by $D_\alpha = c_\alpha \ell_\alpha^*$ for $\alpha = T, P$. Fig. (3) illustrates this property for an infinite beam ($b = -a = +\infty$). It compares the numerical solution of the radiative transfer equation (22) and the analytical solution of the diffusion equation (27) at various times for $k_0 = \gamma^{-1}|k| = 2$, $\eta_\alpha = 0$, and Gaussian correlations of parameters with the same data as before.

For a cylindrical shell, the in-plane/bending modes which are coupled in the radiative transfer regime satisfy an equipartition rule of their energies in the diffusive regime: $w_P(\mathbf{x}, |\mathbf{k}|, t) = w_S(\mathbf{x}, \frac{\varrho}{c_S} |\mathbf{k}|, t)$ for all couples $(\mathbf{x}, \mathbf{k})$ in this regime, and w_P satisfies:

$$\partial_t w_P(\mathbf{x}, |\mathbf{k}|, t) = \nabla_x (D_{PS}(|\mathbf{k}|) \nabla_x w_P(\mathbf{x}, |\mathbf{k}|, t)) , \quad (28)$$

with:

$$D_{PS}(|\mathbf{k}|) = \frac{c_P c_S}{2(c_P^2 + c_S^2)} \left[c_S \ell_P^*(|\mathbf{k}|) + c_P \ell_S^* \left(\frac{c_P}{c_S} |\mathbf{k}| \right) \right] .$$

All of this is described in detail in (Savin 2004).

6 DISCUSSION

The behavior of the solution of the radiative transfer equation (22) at early times is fundamentally different from the behavior of the solution of a diffusion equation (27). The initial wave front is progressively smoothed by the random heterogeneities until it takes the shape of a diffusive halo. Without these heterogeneities, no smoothing can happen and no diffusive regime can be reached. Boundaries play a similar role in that they also contribute to the emergence of a diffusive behavior. In this regime information on the wave direction has been completely lost, as is apparent on Fig. (2) at late times and in Eq. (27) or in Eq. (28), where $\mathbf{k}$ contributes only through its magnitude.

As regards the numerical schemes used to solve the radiative transfer equations, it should be noted that the discontinuous finite element method is quite demanding in terms of computer power and storage. Indeed, both the space and phase dimensions must be discretized. However as the sought solution is rather smooth if random heterogeneities and/or boundaries are taken into account, coarse meshes can be used – at least much coarser than those that would be necessary for the proper discretization of a wave equation at high frequencies. The direct Monte-Carlo method is easier to implement and less demanding in terms of computer resources, but it is also much less versatile.

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